$(0,1)$. *Suppose that* $\{(X_i,T_i)\}_{i=1}^n$ *and* $(X_{\text{test}},T_{\text{test}})$ *are drawn i.i.d and that the censoring times satisfy the conditional independence assumption (Assumption 2.1). Further, assume that there exists a constant* $\gamma_\tau > 0$ *such that the weights satisfy* $w_\tau(x) \le \gamma_\tau$ *for almost all* x*. Then, with probability at least* $1-\delta$ *over the draws of* $\mathcal{D}$*, the LPB* $\hat{L}(x) = \hat{q}_{\hat{\tau}}(x)$ *from* (7) *satisfies*

$$\mathbb{P}\left[T_{\text{test}} \ge \hat{L}(X_{\text{test}})|\mathcal{D}\right] \ge 1-\alpha- \sup_{\tau\in[0,1]}\left\{\sqrt{\frac{2\gamma_\tau^2+5}{|\mathcal{I}_2|}\cdot\log\left(\frac{1}{\delta}\right)}\right\}.$$

Importantly, the above guarantee holds in finite samples, for any LLM $\mathcal{G}(x)$, any prompts distribution P_X, sampling-budget B, and regardless of the accuracy of the quantile estimators $\hat{q}_\tau(x)$. However, this coverage bound depends on the supremum of all possible weights with non-zero probability, denoted by γ_τ. In particular, as γ_τ increases, the bound becomes looser.

In Appendix C, we further analyze the calibration method in greater depth by studying the properties of $\hat{\alpha}(\tau)$ defined in (6). Specifically, in Proposition C.2, we make an illustrative assumption that, for any fixed τ, each time-to-unsafe-sampling T_i is miscovered by $\hat{q}_\tau(X_i)$ at the same constant rate, conditional on the calibration prompts $\{X_j\}_{j\in\mathcal{I}_2}$. This assumption is satisfied by the oracle conditional quantile $q_\tau(\cdot)$ because, by definition, $\mathbb{P}\left[T_i < q_\tau(X_i) \mid X_i\right] = \tau$ for every i, and hence also conditional on $\{X_j\}_{j\in\mathcal{I}_2}$. Under this assumption, we prove that the conditional variance $\text{Var}\left[\hat{\alpha}(\tau) \;\middle|\; \{X_j\}_{j\in\mathcal{I}_2}\right]$ is **linearly monotone** with regard to the mean calibration weight

$$\overline{w}_\tau = \frac{1}{|\mathcal{I}_2|}\sum_{i\in\mathcal{I}_2} w_\tau(X_i). \tag{8}$$

The above discussion shows that any censoring design producing large weights will both weaken the PAC-type coverage bound (through a large supremum weight γ_τ) and increase the variance of the empirical miscoverage estimator (through a large mean weight $\bar{w}_\tau$). Indeed, these two properties guide our `Optimized` censoring design presented in the following section, where we both (i) cap γ_τ by a constant value; and (ii) minimize the mean weight $\bar{w}_\tau$ subject to fully utilizing the sampling-allocation budget B.

5 Proposed Budget-Allocation Mechanism Designs

Having formulated the general calibration scheme and the factors governing its coverage, we first introduce a `Naive` budget allocation to illustrate the inherent challenges of this task. Then, we present our flagship `Optimized` approach, which effectively overcomes the limitations of this naive baseline.

5.1 `Naive` Budget Allocation

As a conceptual warm-up, we introduce a simple baseline budget allocation strategy, the `Naive` approach, which defines each censoring time C_i as a random variable *independent of the prompt* X_i. Under the naive assumption that the LLM outputs are i.i.d, $T_i|X_i$ is a Geometric random variable with infinite support. Hence, we design C_i to be distributed geometrically as well: $C_i \sim \text{Geom}(\min(|\mathcal{I}_2|/B,1))$, $\forall i\in\mathcal{I}_2$, which satisfies the budget constraint in (2) by design. This `Naive` approach is summarized in Algorithm 4 in Appendix B.2.

Under this `Naive` allocation, the event $\hat{q}_\tau(X_i) \le C_i$ can be rare, especially for large $\hat{q}_\tau(X_i)$. This makes $\bar{w}_\tau$ and γ_τ excessively large. Consequently, the resulting coverage rates might deviate from the nominal level—as reflected in our experiments in Section 6. This discussion motivates the method presented below, which utilizes the budget more efficiently and designs the censoring time as a function of the prompt.

5.2 Our Main Proposal: `Optimized` Allocation

To better utilize the calibration budget in the computation of $\hat{\alpha}(\tau)$ in (6), we aim to design prompt-dependent censoring times C_i that maximize $\mathbb{P}\big(\hat{q}_\tau(X_i) \le C_i\big)$. Ideally, we would like to set the censoring times for each τ to increase this probability. However, this is infeasible as we construct the dataset only once. As a result, we must commit to a single budget allocation by choosing a specific quantile level τ to consider. For this purpose, we assume that the quantile estimators are monotone in τ. This assumption can be enforced by sorting $\hat{q}_\tau(X_i)$. Consequently, $\forall\tau_1 \le \tau_2,\ \mathbb{I}\{\hat{q}_{\tau_2}(X_i) \le C_i\} \Rightarrow \mathbb{I}\{\hat{q}_{\tau_1}(X_i) \le C_i\}$. Thus, if we were to naively set $C_i = \hat{q}_1(X_i)$ as the 100%–quantile, we would indeed maximize $\mathbb{P}\big(\hat{q}_\tau(X_i) \le C_i\big)$ for all $\tau\in[0,1]$. However, $\hat{q}_1(X_i)$ can be excessively large, e.g., this 100%-quantile is infinite if $T_i \mid X_i$ follows a geometric distribution. As a result, this naive choice is infeasible under our finite budget constraint $\mathbb{E}\left[\sum_{i\in\mathcal{I}_2} C_i|\{X_i\}_{i\in\mathcal{I}_2}\right] \le B$.

Instead, we consider a more practical compromise: we select a quantile level $\tau_{\text{prior}} \ll 1$ that represents our prior belief $\hat{\tau}\in[0,\tau_{\text{prior}}]$. For example, if the target coverage level $1-\alpha$ is not too extreme (e.g., 90%) and the uncalibrated quantile estimator $\hat{q}_\tau$ is reasonably accurate, a prior level such as $\tau_{\text{prior}} = 20\%$ would be a reasonable choice. Importantly, constraining τ to lie within this prior range affects only the informativeness of the LPB, and not its validity; see Appendix D.6.

After choosing the prior level τ_{prior}, we turn to set a prompt-adaptive censoring time C_i using the estimated quantile $\hat{q}_{\tau_{\text{prior}}}(X_i)$. Following (6), the i-th sample contributes to $\hat{\alpha}(\tau)$ if $C_i \leq \hat{q}_{\tau_{\text{prior}}}(X_i)$ for all $\tau \in [0, \tau_{\text{prior}}]$. Combining this observation with the monotonicity of $\hat{q}_\tau$, a sensible choice for C_i that avoids wasted samplings would either be (i) $C_i = \hat{q}_{\tau_{\text{prior}}}(X_i)$, or (ii) $C_i = 0$. Other choices for C_i that are strictly greater or smaller than $\hat{q}_{\tau_{\text{prior}}}(X_i)$ would result in unnecessary generations that do not contribute to the miscoverage estimator $\hat{\alpha}(\tau)$. Furthermore, to bound γ, we replace the raw quantile estimates $\hat{q}_{\tau_{\text{prior}}}(X_i)$ with a capped variant $\hat{f}_\tau(x) := \min(\hat{q}_\tau(x), M)$, where M is a constant defined by the user. Following this, we define

$$C_i := \text{Ber}(\pi_i) \cdot \hat{f}_{\tau_{\text{prior}}}(X_i), \tag{9}$$

where $\text{Ber}(\pi_i)$ is a Bernoulli random variable with success probability of our choice π_i. We now show how to set this probability to minimize the average weight $\bar{w} = \frac{1}{|\mathcal{I}_2|} \sum_{i \in \mathcal{I}_2} \frac{1}{\pi_i}$ in order to reduce the variance of $\hat{\alpha}(\tau)$. By this design, we get $\mathbb{E}[\sum_{i \in \mathcal{I}_2} C_i | \{X_i\}_{i \in \mathcal{I}_2}] = \sum_{i \in \mathcal{I}_2} \hat{f}_{\tau_{\text{prior}}}(X_i)\, \pi_i$. In turn, we can minimize $\bar{w}$ subject to the budget constraint by solving the following convex optimization problem:

$$\pi^* = \underset{\pi \in [0,1]^{|\mathcal{I}_2|}}{\text{argmin}} \frac{1}{|\mathcal{I}_2|} \sum_{i \in \mathcal{I}_2} \frac{1}{\pi_i} \text{ s.t. } \sum_{i \in \mathcal{I}_2} \hat{f}_{\tau_{\text{prior}}}(X_i)\, \pi_i \leq B. \tag{10}$$

In Appendix B, we introduce Algorithm 8, which efficiently solves the above optimization problem. Proposition B.2 ensures this algorithm returns a unique and strictly positive solution π^* that satisfies the budget constraint (2). Importantly, because the solution π^* depends on the entire set of calibration prompts $\{X_i\}_{i \in \mathcal{I}_2}$, the resulting weights inherit this dependence. To make this explicit, we denote the weight associated with the i-th example as $w(\{X_j\}_{j \in \mathcal{I}_2}, i) = 1/\pi_i^*$. We remark that our coverage guarantee from Theorem 4.1 holds also for this formulation of the weights; see Appendix D.6 for details. Furthermore, these weights are also guaranteed to be upper-bounded by the same expression for γ by trimming the quantile estimates with M:

Proposition 5.1 (Maximal weight bound)**.** *Suppose that* $\max_{i \in \mathcal{I}_2} \hat{f}_{\tau_{\text{prior}}}(X_i) \leq M$*, then, the weights* $w(\{X_j\}_{j \in \mathcal{I}_2}, i)$ *induced by solving* (10) *are upper bounded by* $\gamma = \max(|\mathcal{I}_2| \cdot M/B, 1)$.

All proofs are deferred to Appendix D. The above proposition shows us that weights derived by the solution π^* are bounded by γ. Once all C_i-s are set using the ideal π_i^*, we can use our calibration procedure from Section 4, with the sole modification that the search space for τ is now $\mathcal{T} \cap [0, \tau_{\text{prior}}]$. This `Optimized` calibration scheme is outlined in Algorithm 7 in Appendix B.5.

6 Experiments

We evaluate our methods on both synthetic (Section 6.2) and real (Section 6.3) datasets. For a full description of how we train models to estimate the conditional quantiles of T given X, along with additional experiments, implementation details, and runtime considerations for our calibration methods, see Appendix E. Since there are no established baseline methods in this novel setting, we hereafter introduce two new calibration baselines, `Basic` and `Trimmed`, for comparison. We evaluate our proposed `Optimized` method against these two new baselines, as well as a vanilla uncalibrated quantile regression model and the calibrated `Naive`. The following experiments indicate that our `Optimized` budget allocation leads to more informative LPBs that attain a coverage rate closer to the desired level compared to the baselines.

6.1 Baseline Allocation Strategies

`Basic`. We follow the censoring times design from the `Optimized` method, and set them as in (9): $C_i := \text{Ber}(\pi_i) \cdot \hat{q}_{\tau_{\text{prior}}}(X_i)$. However, unlike the `Optimized` approach, here we determine the success probability using a fixed heuristic: $\pi_i = \min(B/(|\mathcal{I}_2| \cdot \hat{q}_{\tau_{\text{prior}}}(X_i)), 1)$. We then apply the calibration procedure from Section 4 using these generated censoring times. For convenience, Algorithm 5 in Appendix B.3 outlines this `Basic` budget-allocation procedure.

`Trimmed`. We extend the `Basic` budget allocation by trimming the maximal estimated quantile at a fixed threshold $M \in \mathbb{N}$. We do so similarly to the `Optimized` approach, to bound γ. Consequently, the censoring times are now given by $C_i := \text{Ber}(\pi_i) \cdot \hat{f}_{\tau_{\text{prior}}}(X_i)$, where $\hat{f}_\tau(x) := \min(\hat{q}_\tau(x), M)$ and $\pi_i = \min(B/(|\mathcal{I}_2| \cdot \hat{f}_{\tau_{\text{prior}}}(X_i)), 1)$. Following this trimming, the maximum weight $w(x)$ is bounded by $\gamma = \max(|\mathcal{I}_2| \cdot M/B, 1)$. As observed in Theorem 4.1, this allows us to obtain a coverage guarantee that is closer to the desired $1 - \alpha$ level. The rest of the calibration process continues as outlined in Section 4, using the trimmed estimates $\hat{f}_\tau(X_i)$ instead of $\hat{q}_\tau(X_i)$. We summarize this scheme in Algorithm 6 in Appendix B.4.

6.2 Synthetic Data Experiments

We generate a dataset that simulates a high-risk setting with predominantly prompts that can yield unsafe outcomes. To this end, we generate $n = 100{,}000$

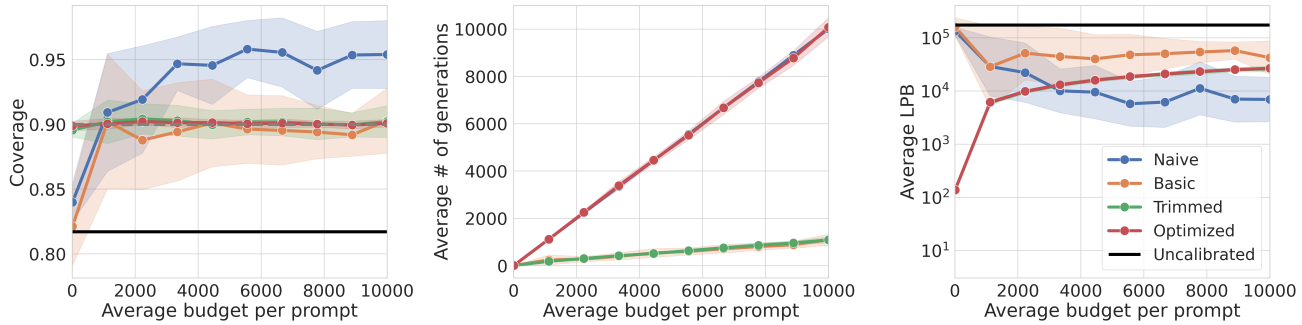


Figure 1: Synthetic experiments. **Left:** Coverage (target 90%). **Center:** Mean number of samplings generated per prompt. **Right:** Mean LPB. Shaded regions represent the standard deviation over 20 runs.

pairs (X_i, p_i), where p_i is the true probability of sampling an unsafe Y_i for $X_i \in \mathbb{R}^d$, with $d = 10$. Specifically, $X_i \mid p_i$, follows a normal distribution, defined in Appendix E.1. The unsafe probability p is chosen to simulate a dataset of "suspicious" prompts, where 90% of which have a high probability for unsafe generation, with the remaining 10% of prompts having a low probability for such an event. Then, we draw $T_i \sim \text{Geom}(p_i)$. We randomly partition the data into training (45%), calibration (45%), and test (10%) sets. For each example in the training set, we generate 500 outputs. We then use this training set to fit the quantile estimators $\hat{q}_\tau(x)$.

We compare (1) the `Uncalibrated` baseline, the raw quantile estimate $\hat{q}_\tau(X)$ at quantile level $\tau = \alpha$; (2) the `Naive` calibration method, which is a slight variant of the method from Section 5.1. The original `Naive` approach tended to be unstable, often yielding trivial LPBs, since $\hat{q}_\tau(X_i) \leq C_i$ holds for very few points at high quantile levels and these samples do not contribute to the miscoverage estimator in (6). To mitigate this, we restrict the τ's search space to $\mathcal{T} \cap [0, \tau_{\text{prior}}]$, as in the adaptive budget allocation methods. We also include (3) the `Basic` method from Section 6.1; (4) its `Trimmed` version from Section 6.1; and (5) our flagship `Optimized` approach from Section 5.2.

Recall that the calibration procedures are inherently random due to the sampling of C_i and the random generations obtained by the model $\mathcal{G}(x)$. Ideally, this randomness should have a small effect on our bounds. To quantify this randomness, we fix the data split, and run each method for $J = 20$ random draws of censoring and time-to-unsafe-sampling times. We index each run by j, and denote the resulting LPB by $\hat{q}_{\hat{\tau}}^{(j)}(X_i)$ and compute:

$$\text{AvgCoverage}^{(j)} = \frac{1}{|\mathcal{I}_{\text{test}}|} \sum_{i \in \mathcal{I}_{\text{test}}} \mathbb{P}\{T_i \leq \hat{q}_{\hat{\tau}}^{(j)}(X_i) | X_i\},$$

$$\text{AvgLPB}^{(j)} = \frac{1}{|\mathcal{I}_{\text{test}}|} \sum_{i \in \mathcal{I}_{\text{test}}} \hat{q}_{\hat{\tau}}^{(j)}(X_i),$$

$$\text{AvgBudget}^{(j)} = \frac{1}{|\mathcal{I}_{\text{test}}|} \sum_{i \in \mathcal{I}_{\text{test}}} C_i^{(j)}.$$

Above, $\mathcal{I}_{\text{test}}$ are the indices of the test points. We report the average and standard deviation of the above quantities over J runs.

Results. We study the effect of the average budget per prompt $B/|\mathcal{I}_2|$ on the coverage and mean LPB obtained by each method. Following Figure 1, we can see that the `Uncalibrated` method does not attain valid coverage. The `Naive` method severely undercovers T for small budgets, and overcovers T for larger budgets. The `Basic` method attains more stable coverage around the desired 90% level under medium-to-high budget, but exhibits high variance. By contrast, the `Trimmed` method attains nearly 90% coverage across all budget constraints. Our flagship `Optimized` calibration method attains tight 90% coverage with minimal variance compared to all other methods. The middle panel highlights that both the `Naive` and `Optimized` approaches fully utilize the budget; however, the latter is tightly regulated around the desired coverage—better utilizing the sampling budget. The right panel demonstrates that the `Trimmed` and `Optimized` methods produce stable LPBs with better statistical efficiency as the average budget per prompt increases. The increased budget allows us to set higher quantile trim values M while enforcing low γ. As the budget increases, the two adaptive methods converge to the performance of the `Basic` technique. However, this technique yields higher LPBs with a higher coverage variability. Additional experiments, including oracle models, varying maximal weights, α levels, and calibration/test splits, are in Appendix E.1.

6.3 Real Data Experiments

We now evaluate the effectiveness of our `Optimized` calibration method on real-world data. For this purpose, we use the RealToxicityPrompts dataset (Gehman et al., 2020), which contains 99,442 sentences designed to measure model toxicity. Each sentence is divided into two halves: the first half serves as the input prompt X_i to a language model, which must then complete the sentence. In this experiment, we generate about 30 million model responses. Because this scale is computationally demanding, we employ a relatively small model, Llama 3.2 1B (Meta AI, 2024; Grattafiori et al., 2024). Furthermore, as we do not have access to human annotations, we rely on the Detoxify-original classifier (Hanu and team, 2020) to label each completion as safe (non-toxic) or unsafe (toxic), using a toxicity score threshold of 0.5.

We randomly split the dataset into training (50%), validation (10%), calibration (20%), and test (20%) sets. On the training set, we sample $N = 500$ responses per prompt and fit a quantile estimator $\hat{q}_\tau(x)$ as detailed in Appendix E. We employ the three calibration methods described in Section 6.2: (1) the `Uncalibrated` baseline, (2) the `Naive` method, and (3) the `Optimized` method, all aiming to attain 90% coverage rate.

In all experiments, we report the empirical mean of the LPBs over the test set. We also estimate the empirical miscoverage by drawing $\min\big(\hat{L}(X_i), 2400\big)$ samples for each prompt. We cap the maximal LLM generations on the test set at 2,400 samplings per prompt due to computational constraints. Since the `Optimized` method trims the quantiles at $M < 2400$ across all evaluated budget levels, by drawing exactly $\hat{L}(X_i)$ samples (with $\hat{L}(X_i) \leq M$) we can compute the miscoverage rate directly. Since $\hat{L}(X_i)$ is unbounded for the `Uncalibrated` and `Naive` methods, we can only report a lower bound on the empirical coverage rate using the 2,400 generations; see also (Gui et al., 2024a). In turn, the empirical lower and upper bounds are given by $\frac{1}{|\mathcal{I}_{\text{test}}|}\sum_{i\in\mathcal{I}_{\text{test}}} \mathbb{I}\big\{\hat{L}(X_i) \leq \tilde{T}_i\big\}$, and $\frac{1}{|\mathcal{I}_{\text{test}}|}\sum_{i\in\mathcal{I}_{\text{test}}} \mathbb{I}\big\{\min(\hat{L}(X_i), 2400) \leq \tilde{T}_i\big\}$, respectively. This evaluation protocol offers a fair comparison under relatively limited computations. To measure the variability introduced by the inherently stochastic calibration procedures, we use the same fixed data split for each run, as in our synthetic experiments.

Results. Figure 2 presents the results across $J = 5$ runs. The `Uncalibrated` baseline overcovers the time-to-unsafe-sampling, resulting in overly conservative LPBs. The `Naive` method, on the other hand, generates invalid LPBs with high variance at small budget levels. In contrast, our `Optimized` method consistently attains near-nominal coverage rates with low variance. Notably, the LPBs produced by the `Optimized` method increase as the budget increases, approaching the values of the `Naive` and surpassing those of the `Uncalibrated` method. Overall, this experiment shows that the `Optimized` method produces LPBs that are both valid across all tested budgets, and become more informative as the budget grows.

7 Discussion

We introduced a novel approach to construct a calibrated LPB for time-to-unsafe-sampling of a given prompt, casting this task as a survival analysis problem. Key to our method is the design of an optimized censoring mechanism that utilizes a given budget constraint on the number of LLM generations. One limitation of our approach is that the coverage guarantee is marginal and does not hold conditionally on a specific subgroup. Consequently, one cannot group prompts with especially low (or high) LPBs and assume that those bounds retain their nominal coverage when conditioned on this selection. In future work, we plan to extend our approach to this conditional inference setting, leveraging the methods by Sesia and Svetnik (2025); Jin and Ren (2025) as a foundation for achieving valid group-conditional coverage. We remark that in our experimental framework, we treat the outputs of the Detoxify audit function as ground-truth events, as obtaining human annotations at the scale of millions of generations is infeasible. This reliance on a surrogate model introduces the potential for label noise. Consequently, a valuable direction for future research is to investigate how training on proxy labels affects the validity of the LPB when evaluated against human-annotated test data, potentially leveraging recent theoretical advancements in conformal prediction with label noise Einbinder et al. (2024). Another limitation is that our calibration assumes the samples are drawn i.i.d. Even in benign real-world setups, the prompt distribution can shift, potentially undermining the validity of our method. A promising direction, which we have already begun to explore, we aim to propose continual or adaptive recalibration mechanisms that adjust to evolving user behavior and emergent risks, possibly by drawing on ideas from (Gibbs and Candès, 2024). Concurrently, we are investigating how to optimally allocate the budget of the training phase to maximize the model's accuracy under limited LLM-sampling resources.

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